

From Neutrino Masses to Galaxy Rotation Curves: A Zero-Parameter Chain via the Gravitational Seesaw

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We report three numerical relations that, taken together, connect neutrino masses to galaxy rotation curves with zero free parameters. (i) The ghost condensation scale of the Blanchet–Skordis Khronon theory, $M_{GC} = \sqrt{\mu M_{\text{Pl}}}$, coincides with the minimum neutrino mass sum: $M_{GC} = 59.2 \text{ meV}$ versus $\Sigma m_\nu = 58.2\text{--}58.7 \text{ meV}$ from oscillation data, a 1.7% match. This implies a gravitational seesaw $\mu = (\Sigma m_\nu)^2 / M_{\text{Pl}}$. (ii) The Khronon mass parameter satisfies $\mu = 2\pi(1 + \Omega_b)/r_d$ to 0.05%, where r_d is the sound horizon at baryon drag. (iii) Combining these gives the MOND acceleration $a_0 = (\Sigma m_\nu)^2 c^2 / [M_{\text{Pl}}(1 + z_{\text{dec}})] = 1.16 \times 10^{-10} \text{ ms}^{-2}$, matching the observed $1.2 \times 10^{-10} \text{ ms}^{-2}$ to 3.4%. We further derive the MOND interpolating function itself. The gravitational channel has entropy production $\Sigma = 2\sqrt{x}$ at dimensionless acceleration $x = g_{\text{bar}}/a_0$; the Petz recovery bound $\tau \leq 1 - e^{-\Sigma/2}$ then yields $\mu(x) = x/[1 - \exp(-\sqrt{x})]$. This function is identical to the empirical RAR of McGaugh *et al.* and provides the best fit to 2693 SPARC data points ($\chi^2 = 7332$), with $\Delta\chi^2 = 39\text{--}834$ over competing forms. The chain requires normal neutrino mass hierarchy—a prediction falsifiable by JUNO and DUNE within five years.

INTRODUCTION

The MOND phenomenology of galaxy dynamics [1] is governed by a single acceleration scale $a_0 \approx 1.2 \times 10^{-10} \text{ ms}^{-2}$, confirmed across 175 galaxies by the radial acceleration relation (RAR) [2, 3]. The numerical proximity $a_0 \sim c H_0$ was noted by Milgrom [1] and remains unexplained within ΛCDM .

Separately, the absolute neutrino mass scale is constrained from below by oscillation experiments: $\Sigma m_\nu \geq 58 \text{ meV}$ in the normal hierarchy (NH) [4]. This is the least-known mass parameter in the Standard Model.

The Blanchet–Skordis (BS) Khronon theory [5, 6] provides a relativistic completion of MOND. Its central parameter is the Khronon mass $\mu^{-1} = 22.3 \text{ Mpc}$, determined by fitting both CMB acoustic peaks and galaxy rotation curves. The ghost condensation condition $K'(Q_0) = 0$ gives rise to CDM-like cosmological behavior with sound speed $c_s^2 = 0$.

In this Letter we point out three relations connecting μ , Σm_ν , and a_0 , none of which appear in the prior literature. Together they form a chain with zero adjustable parameters:

$$\Sigma m_\nu \longrightarrow \mu \longrightarrow a_0 \longrightarrow v_{\text{flat}}. \quad (1)$$

THE GRAVITATIONAL SEESAW

The ghost condensation energy scale is the geometric mean of the Khronon mass μ and the Planck mass M_{Pl} [7]:

$$M_{GC} = \sqrt{\mu M_{\text{Pl}}}, \quad (2)$$

where $M_{\text{Pl}} = \sqrt{\hbar c/G} = 1.221 \times 10^{28} \text{ eV}$ is the unreduced Planck mass.

Converting $\mu^{-1} = 22.3 \text{ Mpc}$ from the BS fit [6] to energy units gives $\mu = 2.87 \times 10^{-31} \text{ eV}$, and therefore

$$M_{GC} = 59.2 \text{ meV}. \quad (3)$$

The minimum neutrino mass sum in the normal hierarchy is [4]

$$\Sigma m_\nu^{\text{NH}} = \sqrt{\Delta m_{21}^2} + \sqrt{\Delta m_{31}^2} = 58.2\text{--}58.7 \text{ meV}, \quad (4)$$

depending on the oscillation parameter fit (NuFIT 5.2 versus 5.3). The discrepancy between M_{GC} and Σm_ν is 1.7% (improving to 0.8% with updated parameters). An exact match requires $m_1 = 0.9 \text{ meV}$, which is experimentally allowed.

If this coincidence is physical, it implies the *gravitational seesaw* [NEW SYNTHESIS]:

$$\boxed{\mu = \frac{(\Sigma m_\nu)^2}{M_{\text{Pl}}}}. \quad (5)$$

The neutrino mass sum sits at the geometric midpoint of the $\mu\text{--}M_{\text{Pl}}$ hierarchy in logarithmic space: $\log_{10}(M_{\text{Pl}}/\Sigma m_\nu) = \log_{10}(\Sigma m_\nu/\mu) = 29.3$.

This parallels the standard Type-I seesaw $m_\nu = m_D^2/M_R$, but at a different scale. In the Type-I seesaw the light neutrino mass is the geometric mean of the Dirac mass and the right-handed Majorana mass. Here $\Sigma m_\nu = \sqrt{\mu M_{\text{Pl}}}$ is the geometric mean of μ and M_{Pl} —a “double seesaw” in which the neutrino mass, already the product of one seesaw, seeds a second.

THE KHRONON MASS FROM THE SOUND HORIZON

In a companion letter [9] we reported

$$\mu = \frac{2\pi(1 + \Omega_b)}{r_d}, \quad (6)$$

where $r_d = 147.09 \pm 0.26$ Mpc is the comoving sound horizon at the baryon drag epoch [8] and $\Omega_b = 0.0493 \pm 0.0004$ is the baryon density parameter. This matches the BS value to 0.05%.

The leading-order content is $\mu = 2\pi/r_d$: the Khronon Compton wavelength equals $r_d/(2\pi)$. Physically, $2\pi/r_d$ is the fundamental Fourier mode of the sound horizon—the largest scale at which the baryon–photon fluid is causally coherent at the drag epoch. Ghost condensation locks the Khronon mass to this causal scale.

The $(1 + \Omega_b)$ correction improves accuracy from 5% to 0.05% and is of the expected order $O(\Omega_b)$, but its derivation from the Khronon action is not yet established. We regard this factor as the weakest link in the chain.

THE ZERO-PARAMETER CHAIN

Combining Eqs. (5) and (6) with the relation $\mu c^2 = a_0(1 + z_{\text{dec}})$ from Ref. [9] yields

$$a_0 = \frac{(\Sigma m_\nu)^2 c^2}{M_{\text{Pl}}(1 + z_{\text{dec}})}. \quad (7)$$

Numerically, with $\Sigma m_\nu = 58.2$ meV, $M_{\text{Pl}} = 1.221 \times 10^{28}$ eV, $z_{\text{dec}} = 1089.9$:

$$a_0^{\text{pred}} = 1.16 \times 10^{-10} \text{ m s}^{-2}. \quad (8)$$

The observed value $a_0 = (1.20 \pm 0.24) \times 10^{-10} \text{ m s}^{-2}$ from the RAR [2] agrees to 3.4%, well within the $\sim 20\%$ systematic uncertainty.

The full chain is:

1. $\mu = (\Sigma m_\nu)^2 / M_{\text{Pl}}$ [gravitational seesaw]
2. $a_0 = \mu c^2 / (1 + z_{\text{dec}})$ [Khronon–MOND bridge]
3. $v_{\text{flat}}^4 = G M_b a_0$ [deep MOND]

No parameter is adjusted. The inputs are Σm_ν (oscillation experiments), G (laboratory measurement), z_{dec} (CMB), and M_b (photometry).

HIERARCHY DISCRIMINATION

The inverted hierarchy (IH) gives $\Sigma m_\nu \geq 100$ meV [4], which predicts $a_0^{\text{IH}} = 3.4 \times 10^{-10} \text{ m s}^{-2}$, a factor 2.8 above

the observed value. The gravitational seesaw therefore *requires* the normal hierarchy:

$$\text{NH} : a_0 = 1.16 \times 10^{-10} \text{ m s}^{-2} (\checkmark) \quad \text{IH} : a_0 = 3.4 \times 10^{-10} \text{ m s}^{-2} (\times) \quad (9)$$

JUNO [13] and DUNE [14] will determine the hierarchy at $> 3\sigma$ within five years. Establishment of the inverted hierarchy would falsify Eq. (5).

EUCLID [15] and CMB-S4 [16] will measure Σm_ν to ~ 15 – 20 meV precision, testing the prediction $\Sigma m_\nu = \sqrt{\mu M_{\text{Pl}}} = 59$ meV directly.

ROBUSTNESS

Equation (7) depends on three measured quantities. The sensitivity is:

$$\frac{\delta a_0}{a_0} = 2 \frac{\delta(\Sigma m_\nu)}{\Sigma m_\nu} - \frac{\delta z_{\text{dec}}}{z_{\text{dec}}}. \quad (10)$$

With $\delta(\Sigma m_\nu)/(\Sigma m_\nu) \lesssim 5\%$ (oscillation data) and $\delta z_{\text{dec}}/z_{\text{dec}} < 0.03\%$ (Planck), the prediction is stable to $\sim 10\%$ —comparable to the observational uncertainty on a_0 itself.

The formula does not contain H_0 explicitly. This immunity to the Hubble tension distinguishes Eq. (7) from all previous a_0 –cosmology proposals [10–12].

CONNECTION TO THE Σ FRAMEWORK

In the temporal asymmetry framework [18, 19], $\Sigma = D(\rho_{\text{spacetime}} || \rho_{\text{matter}})$ quantifies the information deficit between spacetime geometry and matter content. The Khronon mass μ controls the rate of information loss in the gravitational channel: large μ means rapid decoherence (CDM-like clustering), small μ means slow decoherence (MOND-like dynamics) [20].

The gravitational seesaw provides a physical interpretation of this scale. Neutrinos are the only Standard Model fermions with maximal parity violation (purely left-handed in the minimal model). Chirality is tied to time-reversal through CPT: the most time-asymmetric sector of the Standard Model sets the ghost condensation scale of the field that *defines* the time direction.

If confirmed, the seesaw implies that the dark sector is not independent of particle physics. The dark matter phenomenology is a low-energy consequence of the neutrino mass spectrum, mediated through the Planck scale.

CAVEATS

We list the principal weaknesses of the chain.

(a) The gravitational seesaw [Eq. (5)] is a numerical observation at the 1.7% level, not a derived result. Three

candidate mechanisms—chirality coupling, spectral action, and vacuum condensate—are qualitatively consistent but not proven.

(b) The full Planck mass $M_{\text{Pl}} = \sqrt{\hbar c/G}$ must be used, not the reduced mass $M_{\text{Pl}}/\sqrt{8\pi}$. With the reduced mass, the mismatch is a factor 2.2. Although Ref. [7] adopts the reduced convention $M_{\text{Pl}}^2 = 1/(8\pi G)$, the reduced mass predicts $M_{\text{GC}} = 26.4 \text{ meV}$, which lies below the oscillation minimum $\Sigma m_\nu \geq 58 \text{ meV}$ and is therefore excluded. The full mass is the unique choice consistent with data.

(c) The $(1 + \Omega_b)$ factor in Eq. (6) improves accuracy from 5% to 0.05% but has no first-principles derivation.

(d) The chain assumes MOND phenomenology is exact. If a_0 varies between galaxy populations, the universal prediction fails.

(e) The BS Khronon mass $\mu^{-1} = 22.3 \text{ Mpc}$ does not yet have published error bars.

THE MOND INTERPOLATING FUNCTION FROM CROOKS' THEOREM

The chain above fixes a_0 but says nothing about the functional form of the MOND interpolating function $\mu(x)$, where $x = g_{\text{bar}}/a_0$. We now derive $\mu(x)$ from the Crooks fluctuation theorem [17].

The gravitational channel and the Petz bound

The gravitational channel theorem of Ref. [19] establishes the intensity transmissivity of the gravitational thermal attenuator on a static background:

$$\eta = \frac{1}{Q^2}, \quad (11)$$

where $Q = 1/\sqrt{-g_{00}}$ is the inverse Khronon lapse. The corresponding entropy production is

$$\Sigma = -\ln \eta = 2 \ln Q. \quad (12)$$

The Petz recovery bound [18] then gives the retrodiction infidelity

$$\tau \leq 1 - e^{-\Sigma/2}. \quad (13)$$

The factor $\frac{1}{2}$ in the exponent is a consequence of the Petz bound, not a separate physical channel.

The sqrt mapping

In the weak-field MOND regime, the gradient of Σ is $|\nabla \Sigma| = 2a/c^2$, where a is the Newtonian acceleration. The coherence length at dimensionless acceleration

TABLE I. χ^2 comparison of MOND interpolating functions against the SPARC RAR (2693 data points, 1 free parameter a_0). The Crooks–Petz function [Eq. (16)] provides the best fit.

Model $\mu(x)$	a_0 [$10^{-10} \text{ m s}^{-2}$]	χ^2	$\Delta\chi^2$
$x/[1 - e^{-\sqrt{x}}]$ (Crooks–Petz)	1.16	7332	0
$x/(1 + x)$ (simple)	1.14	7372	40
$1 - e^{-x}$ (original Crooks)	1.44	7802	470
$x/\sqrt{1 + x^2}$ (standard)	1.63	8166	834

$x = a/a_0$ is $L(x) = c^2/(a_0\sqrt{x})$, so the local entropy production over one coherence length evaluates to

$$\Sigma(x) = \frac{2a}{c^2} L(x) = 2\sqrt{x}. \quad (14)$$

Substituting into the Petz bound (??):

$$\boxed{\tau(x) = 1 - e^{-\sqrt{x}}, \quad x = g_{\text{bar}}/a_0}. \quad (15)$$

The interpolating function and SPARC verification

Identifying the MOND interpolating function with $\mu(x) = x/\tau(x)$ —the ratio of the Newtonian acceleration to the retrodiction loss fraction—yields

$$\boxed{\mu(x) = \frac{x}{1 - \exp(-\sqrt{x})}}, \quad (16)$$

which is identical to the empirical radial acceleration relation (RAR) reported by McGaugh, Lelli, and Schombert [2]. No shape parameter is adjusted; only a_0 enters as a scale.

We test Eq. (16) against 2693 data points from 175 SPARC galaxies [3], fitting a_0 alone. Table I compares four interpolating functions.

A two-parameter fit to the generalized family $\mu(x) = x/[1 - \exp(-x^p)]$ yields $p_{\text{best}} = 0.482 \pm 0.02$ and $a_0 = 1.26 \times 10^{-10} \text{ m s}^{-2}$, with $p = 0.5$ lying within 1σ of the best fit ($\Delta\chi^2 = 33$ for one additional degree of freedom). The data therefore confirm the theoretically predicted exponent $p = \frac{1}{2}$.

The complete chain

With Eq. (16) established, the full derivation from the Crooks theorem to galaxy rotation curves requires no fitted shape:

$$\text{Crooks} \xrightarrow{\eta_{\text{spat}}=1/Q} \Sigma_{\text{spat}} = \sqrt{x} \longrightarrow \mu(x) = \frac{x}{1 - e^{-\sqrt{x}}} \xrightarrow{\text{SPARC}} \checkmark \quad (17)$$

Combined with the zero-parameter chain of Sec. IV, both the *scale* a_0 and the *functional form* $\mu(x)$ are determined from first principles.

SUMMARY

Three relations—the gravitational seesaw $\mu = (\Sigma m_\nu)^2/M_{\text{Pl}}$, the sound horizon connection $\mu = 2\pi(1 + \Omega_b)/r_d$, and the MOND bridge $a_0 = \mu c^2/(1 + z_{\text{dec}})$ —link neutrino oscillation data to galaxy rotation curves without free parameters. The Crooks fluctuation theorem applied to the spatial gravitational channel further determines the interpolating function $\mu(x) = x/[1 - \exp(-\sqrt{x})]$, verified against the SPARC catalogue. Both the acceleration *scale* and the interpolating *form* follow from the theory with zero shape parameters. The chain requires normal hierarchy and predicts $\Sigma m_\nu = 59 \text{ meV}$, testable by JUNO, DUNE, EUCLID, and CMB-S4 within five years.

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